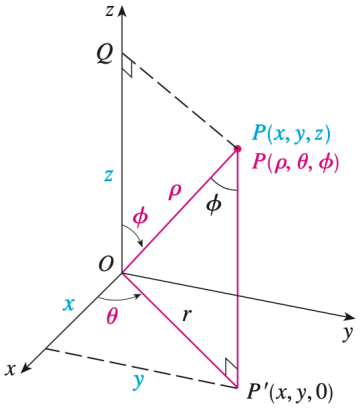
1 Spherical coordinate system

The *spherical coordinates* (ρ, θ, ϕ) of a point P in space are shown below:



Note that $\rho \geq 0$ and $0 \leq \phi \leq \pi$. The spherical coordinate system is especially useful in problems with rotational symmetry about a point in space. The angle ϕ is called the *azimuthal angle* and θ is the *polar angle*.

The relationship between rectangular and spherical coordinates is

$$\boxed{x = \rho \sin\phi \cos\theta,\ y = \rho \sin\phi \sin\theta, z = \rho \cos\phi}.$$

The inverse relation is

$$\boxed{\rho^2 = x^2 + y^2 + z^2, \phi = \cos^{-1}\left(\frac{z}{\rho}\right), \theta = \cos^{-1}\left(\frac{x}{\rho \sin\phi}\right)}.$$

1. (a) The point $(2, \frac{\pi}{4}, \frac{\pi}{3})$ is given in spherical coordinates. Plot the point and find its rectangular coordinates.

   (b) The point $(0, 2\sqrt{3}, -2)$ is given in rectangular coordinates. Find spherical coordinates for this point.

(a) The point $(6, \frac{\pi}{3}, \frac{\pi}{6})$ is given in spherical coordinates. Plot the point and find its rectangular coordinates.

(b) The point $(1, 0, \sqrt{3})$ is given in rectangular coordinates. Find spherical coordinates for this point.

2. Graph the surfaces defined in spherical coordinates by

(a) $\rho = 3$.

(b) $\theta = \frac{\pi}{3}$.

(c) $\phi = \frac{\pi}{3}$.

3. Re-write each equation in spherical coordinates.

   (a) $z^2 = x^2 + y^2$

   (b) $x^2 + z^2 = 9$

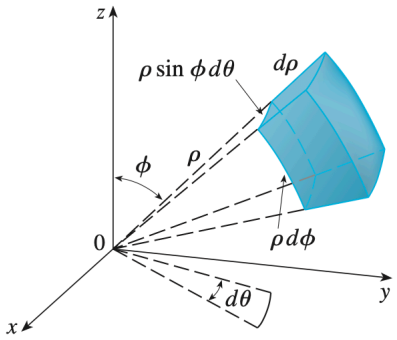
2 Triple Integrals in Spherical Coordinates

To evaluate an integral in spherical coordinates

$$\iiint_E f(\rho \sin\phi \cos\theta, \rho \sin\phi \sin\theta, \rho \cos\phi) dV$$

we need to find the volume element dV. In spherical coordinates, we partition the solid into *spherical wedges*, where a spherical wedge is defined as

$$E = \{(\rho, \theta, \phi) \mid a \le \rho \le b,\ \alpha \le \theta \le \beta,\ c \le \phi \le d\}$$



We see that $dV = \rho^2 \sin\phi \, d\rho \, d\theta \, d\phi$.

3 Triple Integral over a Spherical Wedge

If $f(x,y,z)$ is continuous on the spherical wedge

$$E = \{(\rho,\theta,\phi) \mid a \le \rho \le b,\ \alpha \le \theta \le \beta,\ c \le \phi \le d\},$$

then

$$\iiint_E f(x,y,z)dV = \int_{\phi=c}^{\phi=d}\int_{\theta=\alpha}^{\theta=\beta}\int_{r=a}^{r=b} f(\rho\sin(\phi)\cos(\theta),\rho\sin(\phi)\sin(\theta),\rho\cos(\phi))\rho^2\sin(\phi)d\rho\,d\theta\,d\phi.$$

4. Evaluate $\iiint_B e^{(x^2+y^2+z^2)^{3/2}}dV$, where B is the unit ball $B = \{(x,y,z) : x^2+y^2+z^2 \le 1\}$.

4 Triple Integrals over More General Regions

As before, we can integrate over more general spherical regions of the form

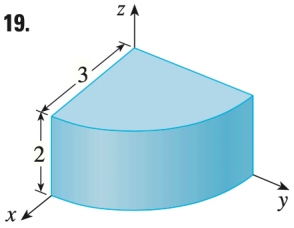
$$E = \{(\rho,\theta,\phi) \mid \alpha \le \theta \le \beta,\ c \le \phi \le d,\ g_1(\theta,\phi) \le \rho \le g_2(\theta,\phi)\}:$$

$$\iiint_E f(x,y,z)dV = \int_{\theta=\alpha}^{\theta=\beta} \int_{\phi=c}^{\phi=d} \int_{\rho=g_1(\theta,\phi)}^{\rho=g_2(\theta,\phi)} f(\rho\sin(\phi)\cos(\theta),\rho\sin(\phi)\sin(\theta),\rho\cos(\phi))\rho^2\sin(\phi)d\rho\,d\theta\,d\phi$$

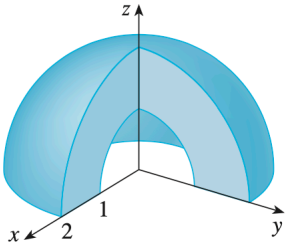
5. Find the volume of the solid that lies above the cone $z = \sqrt{x^2+y^2}$ and below the sphere $x^2+y^2+z^2 = z$.

19–20 Set up the triple integral of an arbitrary continuous function $f(x, y, z)$ in cylindrical or spherical coordinates over the solid shown.

19.



20.



39. $\displaystyle\int_0^1 \int_0^{\sqrt{1-x^2}} \int_{\sqrt{x^2+y^2}}^{\sqrt{2-x^2-y^2}} xy \, dz \, dy \, dx$

17–18 Sketch the solid whose volume is given by the integral and evaluate the integral.

17. $\int_0^{\pi/6} \int_0^{\pi/2} \int_0^3 \rho^2 \sin\phi \; d\rho \, d\theta \, d\phi$

18. $\int_0^{2\pi} \int_{\pi/2}^{\pi} \int_1^2 \rho^2 \sin\phi \; d\rho \, d\phi \, d\theta$